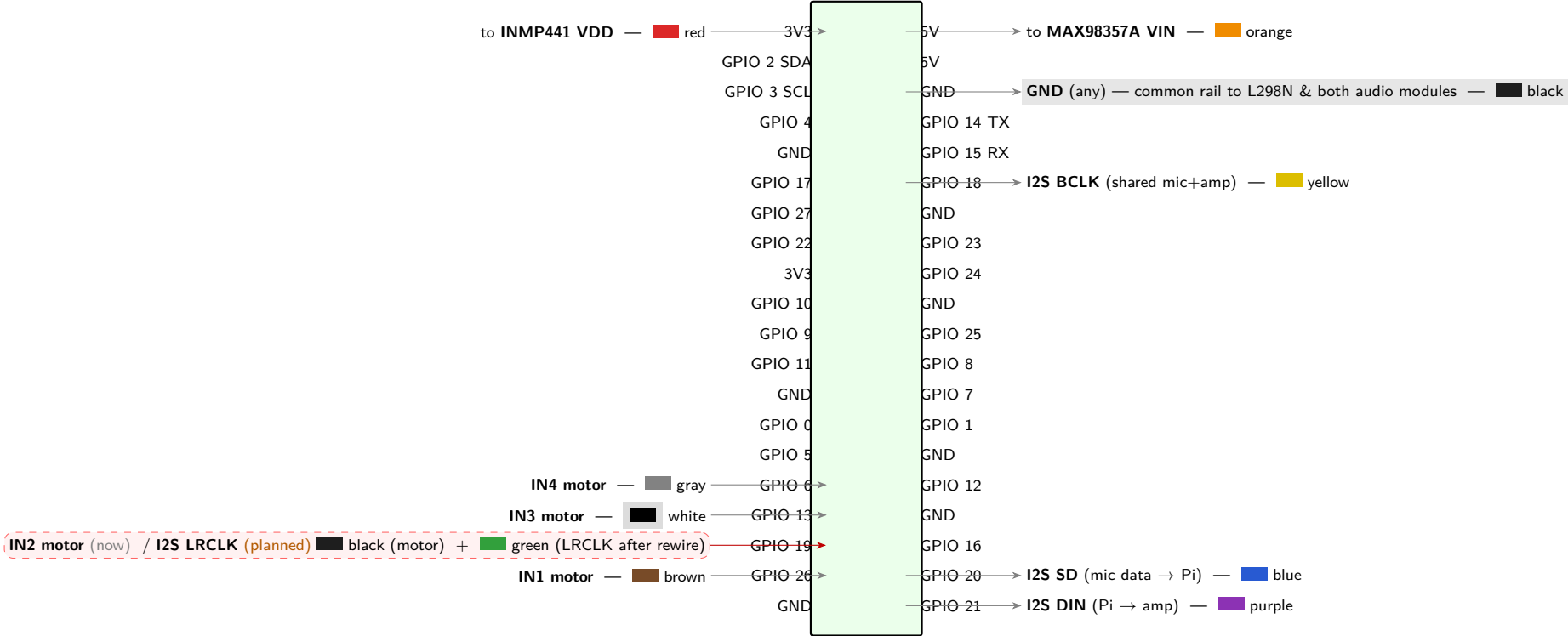


robothector — Raspberry Pi GPIO header & wire-colour reference

Raspberry Pi 40-pin GPIO header (top-down view)



Pin status (header colour)

- currently used (motor IN1–IN4)
- planned for Phase 4 audio
- conflict pin (must rewire IN2)
- power rail (3.3V or 5V)
- ground
- unused / available

Recommended Dupont wire colours

- red** — 3.3V (mic VDD)
- orange** — 5V (amp VIN)
- black** — GND (any)
- yellow** — I2S BCLK (shared)
- green** — I2S LRCLK (shared)
- blue** — I2S SD (mic data)
- purple** — I2S DIN (amp data)
- brown** — IN1 motor (Pi p37)
- black** — IN2 motor (Pi p35)*
- white** — IN3 motor (Pi p33)
- gray** — IN4 motor (Pi p31)

*IN2 wire moves to Pi pin 11 (GPIO 17) when audio is added (frees GPIO 19 for LRCLK).

Connection map

Pi pin	Function	Goes to
1	3.3V	INMP441 VDD
2	5V	MAX98357A VIN
6,9,...	GND	L298N + mic + amp + spk
11	GPIO 17*	L298N IN2 (after rewire)
12	GPIO 18	mic SCK + amp BCLK
31	GPIO 6	L298N IN4
33	GPIO 13	L298N IN3
35	GPIO 19*	mic WS + amp LRC
37	GPIO 26	L298N IN1
38	GPIO 20	INMP441 SD
40	GPIO 21	MAX98357A DIN

* rewire required: free GPIO 19 from motor IN2 by moving IN2 to GPIO 17 (Pi pin 11).